



PISA reading achievement: identifying predictors and examining model generalizability for multilingual students

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Abstract

Reading research in the United States has mainly focused on early or, less frequently, middle grades and on monolingual (MN or English-only) rather than on multilingual (ML) students. To address these gaps, we focused on factors contributing to high school ML students' reading achievement. In particular, we first used machine learning to identify predictors of high school students' reading achievement on PISA 2018. We then conducted multilevel modeling on the entire sample (baseline model) and tested the model's generalizability to ML and MN populations. Results suggest that ML students would benefit from instruction focused on enhancing their reading self-efficacy and increased learning opportunities for extracurricular reading activities. The results also suggest that students, especially ML students, would benefit from schools avoiding grade retention policies and focusing on minimizing truancy and supporting positive peer and teacher relationships. Limitations of the study and future directions are discussed.

Keywords Multilingual students · PISA · Reading · Model generalizability

According to the Organisation for Economic Cooperation and Development (OECD, 2019), reading literacy can be defined as “the active, purposeful and functional application of reading in a range of situations and for various purposes” (p. 28). Reading literacy is not only fundamental for student access to disciplinary knowledge and academic achievement across content areas (e.g., Ardasheva et al., 2021; Shanahan & Shanahan, 2012; Taboada Barber et al., 2015), but also critical for economic growth and public health (Toste et al., 2020). These findings are especially important, both nationally and globally, for multilingual students (MLs), the fastest-growing, yet still understudied student population (e.g., August & Shanahan 2006; Hussar et al., 2020). Reading research in the United States (U.S.), in particular, has focused mainly

on early or middle (less frequently) grades and primarily on monolingual students (MNs; August & Shanahan 2006). This is of concern as while in early/middle grades all learners are still developing their literacy skills and are instructionally supported, high school ML students may be the most at-risk student population (OECD, 2019; Vaughn et al., 2019). In particular, many high-school ML students are still developing their reading ability in the language of schooling and testing, have limited time in school to remedy reading problems, and direct literacy instruction may no longer be available to them at this career-readiness-focused educational level to support their learning needs (OECD, 2019; Vaughn et al., 2019). We recognize that ML students represent a heterogeneous group of students who are often referred to in the literature as English learners or as limited English proficiency, language minority, or culturally and linguistically diverse students. Reflecting the terminology of the most recent English language development standards adopted by over 40 U.S. states and territories (WIDA, 2020), we chose to use the term ML to refer to students who are learning English as an additional language.

Considering recent conceptualizations of reading literacy as being impacted by a host of student-, classroom-, and school-level characteristics (OECD, 2019), empirical evidence supporting such conceptualization (Ardasheva & Tretter, 2013; Feng et al., 2019; Giambona & Porcu, 2015), and the high cost of interventions along with the documented differential impacts of interventions on MN vs. ML students (e.g., Llosa et al., 2016; Vaughn et al., 2015; Vaughn et al., 2019), there is a pressing need to uncover the underlying impacts of multiple, nested systems on improving reading literacy for high school students. This is particularly important because while reading research on high school MLs is limited, evidence from studies focused on younger readers suggests that the prominence of reading comprehension predictors and reading behaviors may differ for ML as compared to MN readers (Frid & Friesen, 2020; Jiménez et al., 1996; Melby-Lervåg & Lervåg 2014). Across age groups, the social aspects of schooling that impact achievement may also differ for MLs (Zaff et al., 2021) as they may experience discrimination, segregation, cultural stereotyping, and differential implementation of programs (Ardasheva et al., 2012; Datnow et al., 2005; Lukes, 2014; Suárez-Orozco & Suárez-Orozco, 2009; Tenenbaum & Ruck 2007).

Identifying and better understanding instructionally malleable factors supporting reading literacy and learning from reading across nested systems is especially important in upper grades “where the language demands of texts are higher and, at the same time, texts are often the primary instructional tool” for many teachers across disciplines (Ardasheva et al., 2021, p. 293), yet not many high school teachers are prepared to support literacy skills development. A focus on MLs is particularly important provided continually documented achievement—including reading and reading subskills achievement (e.g., August & Shanahan 2006; Townsend et al., 2020; Uccelli et al., 2015)—gaps for this important student population.

Therefore, in this study, we use machine learning feature selection and multilevel modeling to explore and identify predictors of student achievement in reading. We leveraged thousands of variables at all levels (student, teacher, school) from the U.S. 2018 Programme for International Student Assessment (PISA) data—the latest PISA assessment with a primary focus on reading—and examined factors that support reading literacy of high school students, with a focus on ML students. Machine

learning feature, in particular, is well suited to work with large-scale data by allowing to explore robust patterns in the data, which could lead to theory confirmation or inductive theory building (e.g., Choudhury et al., 2018; Nelson, 2020). That is, machine learning methods make fewer *a priori* assumptions about the underlying model representing the data best, which allows for replicable identification of patterns (Choudhury et al., 2018). In content analysis, for example, machine learning is used as a tool within a three-step computational grounded theory framework (Nelson, 2020). The three steps, which inform our methodological approach, to some extent, include (a) *pattern detection* using machine learning, (b) *pattern refinement* through “an interpretive engagement with the data” (p. 3) by the researchers, and (c) *pattern confirmation* through further computational or statistical techniques. In other words, machine learning methods act “as a complementary tool” (p. 1) not only for traditional causal inference methods but also for inductive theory-creating methods such as grounded theory (Choudhury et al., 2018).

Recognizing that ML students negotiate multiple academic and social contexts influencing their learning, our study seeks to examine the impacts of multiple layers of social contexts on their reading achievement within an ecological model (Bronfenbrenner, 1977) framework. Below, we describe student-, teacher-, and school-level variables linked to reading achievement by previous studies, after a brief discussion of the PISA assessment and Bronfenbrenner’s ecological model, the main data source, and the theoretical grounding for the present study. We use machine learning to detect and refine patterns at each—student, teacher, and school—level, then use multilevel modeling to confirm these patterns across levels. We finally test the generalizability of these patterns for ML vs. MN students.

Study background and conceptual framing

“PISA 2018 marks the third time that reading is a major domain and the second time that the reading literacy framework receives a major revision” (OECD, 2019, p. 22). Reading is no longer considered an ability acquired early by young children. Rather, it is viewed as an expanding set of knowledge, drawing on resources, dispositions, attitudes, motivation, and strategies that individuals build across contexts, including interactions with parents, peers, teachers, and the community. In other words, reading does not occur in a vacuum (McCrudden & Schraw, 2007; Snow, 2002). Thus, PISA 2018 considers a host of constructs associated with reading literacy at the student, teacher, and school levels.

Aligned with PISA 2018 framework, our study was guided by Bronfenbrenner’s (1977) ecological systems theory. The theory considers factors affecting learning across multiple nested systems, ranging from the *individual* (personal characteristics) and the *micro-* and *meso-systems* (the immediate learning contexts and the interface among them) to the *exosystems* (larger contexts affecting learning indirectly; Crawford et al., 2020). In particular, our study drew on three of the four key elements central to Bronfenbrenner’s Process–Person–Context–Time (PPCT) model (Bronfenbrenner & Morris, 1998). Aligned with the multiple nested systems and the PISA 2018 reading literacy Framework (OECD, 2019), the three components of the PPCT

Table 1 Bronfenbrenner's ecological systems theory aligned to PISA 2018 Assessment and Analytical Framework

<i>Bronfenbrenner's Ecological Systems</i> (Bronfenbrenner & Morris, 1998; Crawford et al., 2020)	<i>Sample Characteristics from PISA 2018 Reading Literary Assessment and Analytical Framework</i> (OECD, 2019)
Person, the individual's unique set of characteristics: · <i>demand characteristics</i>, characteristics eliciting individuals-within-the-environment reactions such as language status; · <i>resource characteristics</i>, resources available to individuals such as experience; · <i>force characteristics</i>, dispositions such as motivation and perceptions Process, individual-immediate environment interactions Context, the higher-level nested settings	Student background and non-cognitive/metacognitive constructs: · <i>general background</i> such as family and home background · <i>reading related background</i> such as out of school reading experience · <i>non-cognitive/metacognitive constructs</i> such as attitudes, motivation, strategies Teaching and learning constructs, (proxies of teacher responsiveness to student needs), teacher preparation, practices for reading, curriculum School policies and governance constructs, environment for reading, climate, context, and routines

Note. PISA = Programme for International Student Assessment

model central to our study include *person*, conceptualized as an individual's unique set of characteristics; *process*, conceptualized as individual-environment interactions within an immediate learning context such as a classroom; and *context*, conceptualized as a higher-level nested setting such as school.¹ From this theory's perspective, a learning outcome is a function of an individual's characteristics and processes in the immediate (proximal) and larger (distal) environmental contexts (Bronfenbrenner & Evans, 2000). It's important to note that the framework positions 'process' in the first place as "proximal processes are posited as the primary engines of development" (Bronfenbrenner & Morris, 2006, p. 798). Yet, in alignment with PISA reading framework (OECD, 2019), we position 'person' first. This decision is also grounded in Bronfenbrenner and Morris' (2006) argument that while person's cognitive and socioemotional characteristics are treated as dependent variables by most research, such characteristics are far less often examined "as precursors and producers of later development. From the perspective of the bioecological model, their effectiveness in the latter role derives from their capacity to influence the emergence and operation of proximal processes" (p. 810). Table 1 describes the alignment of the model's person, process, and context components to the PISA 2018 Assessment and Analytical Framework (OECD, 2019).

The ecological systems theory has been used across age groups and learning domains (e.g., Ardasheva & Tretter 2013; Carhill et al., 2008; Eamon, 2001; Far-

¹ It is important to note that the length of time the individual has been exposed to these immediate and distal processes and settings and the interactions among process, person, and context factors are also part of the theory. Considering these impacts and their interactions over time, however, is outside the scope of our paper due to the cross-sectional nature of the data collected by PISA.

rant & Zubrick, 2012). Ponitz et al. (2009), for example, found a complex pattern of relationships among individual and environmental factors predicting early reading achievement. In particular, classroom quality contributed to students' achievement via its effects on engagement, controlling for students' family socio-demographic risk and prior achievement. In a study of adolescent MLs, Suárez-Orozco et al. (2008) found that a student-centered model (and prior to accounting for English proficiency, maternal education, and student behavioral engagement) accounted for about 10% of the variance in reading achievement. Analyses testing the school-centered model revealed that all school-quality variables examined—percentage of diverse students and average English Language Arts proficiency, poverty, and attendance rates—were significant predictors of reading achievement.

Combined, these findings highlight the importance of examining both individual and environmental factors associated with learning. The constructs central to our study aligned with Bronfenbrenner's person, process, and context ecological systems constructs (Bronfenbrenner & Morris, 1998) are operationalized in the present investigation as: student characteristics (e.g., demographics, non-cognitive and meta-cognitive factors), teacher characteristics (e.g., preparation, job satisfaction; proxies of classroom experiences and teacher responsiveness to student needs), and school characteristics (e.g., policies, climate). We further outline these predictors below.

Person: student-level predictors

Research suggests that student background characteristics such as SES, parental education, prior learning, migration status, and reading experiences beyond school may be important student-level predictors of reading outcomes (e.g., Dewan 2016; Ponitz et al., 2009; Carhill et al., 2008). Giambona & Porcu (2015), for example, found that educational resources available to students at home, such as desks, computers, quiet places to study, and reference books, were important predictors of achievement for lower-readers. At the same time, the number of books at home was more important for higher-readers. Higher occupational status and educational levels of parents were predictive of reading achievement regardless of students' SES, "supporting inter-generational dependence of educational achievement" (Giambona & Porcu, 2015, p. 102; see also Carhill et al., 2008; Valenzuela et al., 2015). Yet, the positive effects of parental occupational status and educational levels were lower for lower-achieving students (Giambona & Porcu, 2015). Such variations suggest a need to extend this research to compare these relationships by student language status, as even highly educated parents of ML students may not be able to provide adequate support for their children reading in an additional language.

Similar to student background characteristics, many empirical studies link higher reading outcomes with such non-cognitive and metacognitive *individual* factors including self-efficacy, metacognition, motivation, and engagement, among both ML and MN students (e.g., Ardasheva 2016; Cockroft & Atkinson, 2017; Meng et al., 2016; Okumura, 2014; van Gelderen et al., 2004). For example, Kirsch et al. (2002) found that the relationship between reading and engagement was larger than that between SES and reading. Guthrie et al. (1999) found that, controlling for self-efficacy and prior reading achievement/topic knowledge, motivation significantly pre-

dicted reading amount. Synthesizing findings from 132 articles, Toste et al. (2020) found a statistically significant, moderate relationship between motivation and K-12 students' reading. Similarly, a substantial body of work documented significant relations between strategy use (metacognitive factors) and reading comprehension (e.g., Ardasheva & Tretter 2013; Muijselaar et al., 2017; Schoonen et al., 1998).

Studies also show that instruction focused on non-cognitive and metacognitive factors enhances reading outcomes (Ardasheva et al., 2017; Pressley, 2000). These include teacher scaffolding, stimulation, and “support for autonomy, competence and ownership. . . are powerful ways of improving students' reading proficiency, awareness of strategies (metacognition) and engagement (motivation) in reading” (OECD, 2019, p. 236; see August & Shanahan, 2006; National Reading Panel 2000; Okumura, 2014). Although the current curriculum for adolescents may not include direct reading instruction (OECD, 2019), students still experience disciplinary literacy practices across subject areas (i.e., reading across disciplines/disciplinary genres)—in the U.S., this may be fueled by such reform documents as Common Core State Standards (National Governors Association Center for Best Practices & Council of Chief State School Officers, 2010)—and may greatly vary on their levels of self-efficacy, motivation, engagement, and metacognitive skills.

Overall, student background characteristics, non-cognitive and metacognitive factors, as well as their immediate learning environments are important to explore as they can affect the extent to which students respond to reading interventions (Vaughn et al., 2019). This is particularly important to consider in light of student language status, as such factors as self-efficacy or engagement and their impacts across learning outcomes may vary for ML vs. MN students (e.g., Taboada Barber et al., 2015; Baroody et al., 2014; Griggs et al., 2013). This variance in self-efficacy or engagement could potentially influence the ways ML students interact with their teachers, a group to which we now turn.

Process: teacher-level predictors

Similar to student characteristics, teacher-level factors such as preparation, self-efficacy, and years of experience have also been found to impact student learning. In a meta-analysis of meta-analyses, Hattie (2009, as cited in Kim et al., 2019), lists over 250 factors impacting student achievement (the full upgraded list can be found at <https://visible-learning.org/wp-content/uploads/2018/03/VLPLUS-252-Influences-Hattie-ranking-DEC-2017.pdf>). These are categorized under seven sources of impact, with teacher (e.g., teacher education and attributes) and teaching (e.g., instructional strategies) occupying a prominent role. Effective teacher preparation, in particular, plays a significant role in achieving the desired learning outcomes in the classroom. The disconnect between undergraduate education and teacher preparation programs generates many obstacles that undermine teacher practices and self-efficacy in implementing classroom instruction (Faez & Valeo, 2012; Enchandia et al., 2007). Feng et al. (2019), for example, found that teachers who completed literacy-specific coursework during preparation and who received discipline-specific mentoring had higher self-efficacy. Teacher personality and emotions have also started attracting researchers' attention in recent years (Chen, 2019; Kim et al., 2019; Uitto et al., 2015).

Teacher emotions intertwine with motivation and cognition and can be influenced by such factors as professional relationships, school environments, and job satisfaction (Kim & Schallert, 2011; Uitto et al., 2015; Yin et al., 2017), all impacting instructional behaviors (e.g., Saunders 2013; Trigwell, 2012). Empirical studies have documented that teachers with positive emotions are more likely to produce an effective learning environment and thus facilitate student learning and development (Yan et al., 2011). However, these findings are often generalized to *all* students, whereas the impacts of such relationships on ML students in particular have not been studied in depth. Finally, while students and teachers interact with one another within the classroom, they also interact outside the classroom within the bounds of the school itself, walking through hallways, engaged in clubs, and participating in larger, schoolwide programming. We now turn to this final level of predictors.

Context: school-level predictors

School-level factors reflecting “norms, goals, values, interpersonal relationships, teaching and learning practices, and organizational structures” (National School Climate Council, 2007, p. 5) affect the life and learning of students. Indeed, school effects on the achievement of both ML and MN students are well documented (e.g., Fry 2008; Jehangir et al., 2015; Kao & Thompson, 2003; Ruus et al., 2007; Thapa et al., 2013). For example, Suárez-Orozco et al. (2008) found that the percentage of low-SES students in a school along with the schools’ racial representation, the average attendance rate, and the average success rate (the percent of students scoring proficient on the state-mandated language arts test) explained about 32% of the variance in reading test scores of ML students. Similarly, the average school SES level significantly predicted reading achievement in Okumura (2014). Notably, the school-level mean of SES (and that of teacher engagement stimulation) explained more variance in student reading than did student-level predictors – highlighting the potential of school-level variables exerting influence on learning. That said, it is worthwhile noting that Okumura’s study used a limited set of variables at the student level, no variables at individual teacher level, and the analyses were not disaggregated by students’ language status. This is unfortunate, and indicated that attending schools with lower SES and achievement profiles may have a negative impact on ML students (see also Ardasheva et al., 2012; Carhill et al., 2008; Yoko, 2007). There is further evidence to suggest that relationships between reading achievement and student background characteristics can be impacted by school funding and curriculum (Jehangir et al., 2015). Yet, a trend common to ML populations is to find themselves in schools with “high levels of students living in or near poverty” (Fry, 2008, p. i) as well as in schools that are less safe and supportive (or perceived as such; Sanders et al., 2018; Watkins & Melde, 2009). In turn, schools with poor supportive systems tend to have lower levels of achievement, among other negative outcomes (Astor et al., 2009). By including student-, teacher-, and school-level variables, we hope to uncover a more nuanced understanding of the factors influencing student reading performance.

The present study

Multilingual students, a group historically underrepresented in higher-paid jobs (Fraleigh-Lohrfink et al., 2013), often need to negotiate multiple academic and social contexts that influence their learning across disciplinary domains. Research is needed on curricular equity and access to understand how academic success of this important student population is affected by background and instructional factors, educational policy, school structures, and “the relational aspects of schooling” across content areas (Callahan, 2018, p. 26).

This is particularly important for reading literacy, foundational for success across subject areas. Therefore, we examined the impacts of multiple layers of individual and social contexts on student reading outcomes, starting with selecting predictors and generating a baseline model using the entire PISA data set, then examining the model’s generalizability for ML vs. MN students. The following research questions guided our work:

1. What predictors of reading achievement are most salient at the student, teacher, and school levels?
2. What are the relative contributions of the most salient student-, teacher-, and school-level constructs to students’ reading achievement?
3. What are the relative contributions of the most salient student-, teacher-, and school-level constructs to students’ reading achievement disaggregated by language status?

Method

Data and participants

Data sources

Participants. The U.S. 2018 PISA reading assessment sample data and the accompanying student (including home and parent), teacher, and school questionnaires were used for the analysis. The sample included 4,838 U.S. students aged 15–16 years old (49.1% female) and 3,526 teachers (50.6% female; 29.8% reading or language-specific teachers) from 164 schools (6.8% private). Students were 10th (74.4%) or 11th graders (17.1%). ML sample included 736 (52.6% female) students from 126 schools and that of MN students was 4054 (48.7% female) students from 163 schools. Most of the ML students in the sample reported speaking Spanish at home ($N=517$). The average cluster size (i.e., number of students per school) was 30 for the entire sample (ranging from 1 to 39), 25 for the MN sample (ranging from 2 to 39), and 6 for the ML sample (ranging from 1 to 22). There were 48 students (1%) who did not report their language status and were excluded from the study.

Of the 4,838 students in the entire sample, 43.3% reported their race/ethnicity as White, not Hispanic ($N=2,097$), 15.9% as Black or African American ($N=769$),

25.2% as Hispanic or Latino ($N=1,221$), 5.9% as Asian ($N=284$), 7.6% as two or more races ($N=367$), and 1% as other ($N=47$). There were an additional 1.1% of the students ($N=53$) who did not report their race/ethnicity information. Of the 4,054 students in the MN sample, 50.5% self-reported as White, not Hispanic ($N=2,047$), 18.3% as Black or African American ($N=741$), 17.2% as Hispanic or Latino ($N=698$), 3.8% as Asian ($N=153$), 8.8% as two or more races ($N=356$), and 1.1% as other ($N=43$). The non-response rate was 0.4% ($N=16$). Of the 736 ML students, 8% self-reported as White, not Hispanic ($N=43$), 3.5% as Black or African American ($N=26$), 70.8% as Hispanic or Latino ($N=698$), 17.8% as Asian ($N=131$), 1.4% were two or more races ($N=10$), and 0.5% as other ($N=4$). The non-response rate was 0.1% ($N=1$). Additionally, PISA data included an index of student economic, social and cultural status (ESCS) - a standardized composite variable computed from student background variables such as home possession and parent education (OECD, 2019). Statistics showed that the ML student sample had a lower ESCS status ($M=-0.61$, $SD=1.12$) on average as compared to the MN sample ($M=0.19$, $SD=0.93$). The mean ESCS status for the entire student sample was 0.06 ($SD=1.01$).

The reading/language-specific teachers included in the data had an average of 14.31 years of teaching experience ($SD=9.17$). The vast majority of the teachers (97.4%) reported as full-time (more than 90% of full-time hours). The non-response rate for teachers was 1% ($N=10$).

Variables and measures

Reading outcome. The outcome variables were the 10 sets of plausible values (PVs) of student reading performance. PVs, proxies of performance drawn from students' proficiency distributions, are typically used in large-scale assessment programs to account for the estimation bias due to the Booklet design (Beaton et al., 2011; Braun & von Davier, 2017). For these reasons, it is recommended to use all sets of PVs for the analysis (Braun & von Davier, 2017; Rutkowski et al., 2010). Table 2 provides (unweighted) descriptive statistics. An independent samples *t*-test was conducted to examine the mean reading performance differences between ML ($M_{ML}=482.16$, $SD_{ML}=106.45$) and MN students ($M_{MN}=510.89$, $SD_{MN}=107.28$). The analysis was conducted by taking into account all the PVs and the student sampling weights. Results suggested a significant difference in average reading scores across the two groups, $\Delta M=28.73$, $t=27.88$, $p<.001$, Cohen's $d=0.18$.

Contextual factors. The 2018 PISA questionnaire data included a total of 1482 variables across all levels, of which 943, 347, and 192 were at the student, teacher, and school levels, respectively. Due to space limitations, only the variables identified as predictors in the present analysis are discussed in the paper. A complete variable list can be found on the PISA website (<https://www.oecd.org/pisa/data/2018database/>).

Data analysis

To address the research questions, a three-step analysis strategy was applied in the present study. The three steps included: (1) ranking all the contextual variables collected by PISA based on their statistical contribution to student reading performance

Table 2 Average reading performance across subsamples and PVs

Variables	Entire Student Sample (N=4838)		Multilingual Students (N=736)		Monolingual Students (N=4054)	
	Mean	SD	Mean	SD	Mean	SD
PV #1	500.15	108.45	419.00	97.97	504.66	108.32
PV #2	500.79	107.95	418.16	97.29	505.52	107.66
PV #3	500.30	107.90	422.17	104.56	505.18	107.47
PV #4	501.10	108.48	429.32	102.04	505.09	108.04
PV #5	500.48	108.08	425.51	97.73	505.01	107.59
PV #6	500.95	108.19	421.06	89.97	505.70	107.57
PV #7	499.89	107.71	423.95	104.92	504.47	107.54
PV #8	500.30	108.11	424.37	100.68	504.89	107.86
PV #9	501.08	107.43	419.85	98.44	505.66	107.22
PV #10	500.63	107.93	416.17	94.91	505.36	107.48

Note. PV=Plausible values of student performance in PISA reading data. In PISA 2018 where reading was the major domain, student reading performance was reported on a scale with a mean of 500 and a standard deviation of 100.

using Elastic Net analysis; (2) selecting the most salient predictors supported by the results of the Elastic Net analysis and the literature; and (3) conducting multilevel modeling (MLM) analysis to explore and interpret the relationship between selected predictors and student performance on PISA reading. Below, we describe each step in greater detail.

Step 1: Preliminary and elastic net analyses

Coming from the penalized regression family, the Elastic Net analysis is a machine learning feature (or variable selection technique) used to identify meaningful factors out of a large number of predictors for the outcome (Murphy, 2012). The Elastic Net is a penalized (also known as regularized or shrinkage) regression approach (Bruce et al., 2020; James et al., 2013). The method applies penalty (or regularization) to the regression model in a way that variables with relatively small contributions to the model will have coefficients approaching zero. That is, if an explanatory variable's contribution is only minor in accounting for the variance of the outcome, its coefficient will be shrunk toward zero. The penalty can be applied in different ways (James et al., 2013), two are pertinent to our study. The first is the L1 penalty which applies regularization by penalizing the sum of absolute values of the regression coefficients. The penalized regression with L1 penalty only is known as *Lasso*. The second is the L2 penalty which applies constraints by penalizing the sum of squared coefficients. The penalized regression with L2 penalty only is known as *Ridge*. The Elastic Net method applies both L1 and L2 penalties in the same model making it superior to other approaches (Zou & Hastie, 2005). Another advantage of this approach lies in its robustness in the presence of small sample sizes—even under situations when the sample size is much smaller than the number of the variables (Luan et al., 2020; Zou & Hastie, 2005). The Elastic Net analysis has been widely applied in STEM fields and is now gaining attention in educational research (Goretzko & Bühner, 2020; Murphy, 2012; Yoo, 2018). In the current study, the Elastic Net analysis was accom-

plished using the packages **caret** (Kuhn, 2021) and **glmnet** (Friedman et al., 2021) in R (R Core Team, 2021).

Prior to the analysis, nominal variables were dummy coded. Further, significance tests are not reliable under penalized regressions and usually are not reported. Therefore, all the variables were standardized so that their relative statistical contribution to the outcome could be examined. The proportion of missing data ranged from none to 100% and variables with a missing proportion above 20% were excluded from the analysis (Peugh & Enders, 2004). For the remaining variables, the predictive mean matching (PPM) from the **mice** package (van Buuren & Groothuis-Oudshoorn, 2021) was used to impute the missing data. All sets of PVs were used, resulting in 10 sets of results that served the purpose of variable selection. The overall student sampling weight was also included.

Step 2: Variable selection

In this step, a final list of the variables was decided through synthesizing the results of the Elastic Net analysis in consultation with theoretical and empirical considerations from the existing literature. In practice, it is usually not recommended to interpret coefficients of the variables from the Elastic Net analysis because the coefficients are biased due to the penalization or regularization nature of the approach (i.e., a solution for better performance of the predictive model at a cost of introducing bias to the parameters). Relatedly, and as mentioned earlier, significance tests for the estimated coefficients are not reliable and thus not reported. Because of this, the results of the Elastic Net analysis are usually interpreted by ranking the statistical importance of the variable based on the magnitude of their coefficients. In the current study, the 10 sets of PVs in the PISA data yielded 10 sets of results. The variable rankings varied across the data sets (e.g., a top-ranked variable for one PV set might get a lower ranking for another PV set), to some extent. To arrive at the final variable selection, our study relied on a combination of criteria. First, as noted above, we ranked the variables in terms of their coefficient values for all 10 sets of the results. Next, a panel of project team members with expertise in language learning, literacy, and language assessment examined all sets of ranked variables. The panel focused not only on statistically important variables but also on those with relatively smaller coefficients but strongly supported in the literature as salient contributors to reading. The results were discussed by the panel until an agreement was achieved on the final variable selection. The selected variables were then examined using multilevel modeling (the next step below) in terms of their relative, statistical contributions to reading achievement at each level.

Step 3: Multilevel modeling

At this step, a three-level MLM² was applied to explore and interpret the relationships between the selected variables and student reading performance accounting

² While we originally planned a three-level (student, teacher, school) MLM, a closer examination of the data suggested a two-level model, from a statistical point of view. That is, while students in the PISA

Table 3 Multilevel models**Model 0 – The null model**

$$\text{Level 1: } \textit{Reading}_{ij} = \beta_{0j} + R_{ij}$$

$$\text{Level 2: } \beta_{0j} = \gamma_{00} + U_{0j}$$

Model 1 – The model with student variables only

$$\text{Level 1: } \textit{Reading}_{ij} = \beta_{0j} + \beta_{1j} (\textit{StudentVariables}_{ij}) + R_{ij}$$

$$\text{Level 2: } \beta_{0j} = \gamma_{00} + U_{0j}$$

Model 2 – The model with student- and Teacher variables

$$\text{Level 1: } \textit{Reading}_{ij} = \beta_{0j} + \beta_{1j} (\textit{StudentVariables}_{ij}) + R_{ij}$$

$$\text{Level 2: } \beta_{0j} = \gamma_{00} + \gamma_{01} (\textit{TeacherVariables}_j) + U_{0j}$$

Model 3 – The model with all selected variables

$$\text{Level 1: } \textit{Reading}_{ij} = \beta_{0j} + \beta_{1j} (\textit{StudentVariables}_{ij}) + R_{ij}$$

$$\text{Level 2: } \beta_{0j} = \gamma_{00} + \gamma_{01} (\textit{TeacherVariables}_j) + \gamma_{02} (\textit{SchoolVariables}_j) + U_{0j}$$

for the hierarchical nature of the data. The analysis was first conducted on the entire sample of students (baseline model), and then on the ML and MN student samples, allowing for the investigation of the model generalizability across student populations disaggregated by language status. All the MLM analyses were run using the R **lme4** package (Bates et al., 2021). In the analysis, all sets of PVs were used and the results were then compiled using a multiple strategy procedure for interpretation using the R **mitml** package (Grund et al., 2021).

Prior to the analysis, the fully conditional specification (FCS) method was used to impute the missing data (Grund et al., 2018; Van Buuren 2018) with the R packages **mice** (van Buuren & Groothuis-Oudshoorn, 2021) and **miceadds** (Robitzsch & Grund, 2021). Following Rabe-Hesketh & Skrondal (2006), the sampling weights were included and scaled before the analysis. All student-level variables were group-mean centered while the teacher- and school-level variables were grand-mean centered. Distributions of residuals were examined to evaluate the assumptions of linearity, normality, independence, and homoscedasticity, and no concerns were detected.

When conducting the main MLM analysis, a *null or unconditional random intercept model* with no predictors was first conducted and the intraclass correlation (ICC) was examined for the percentage of variance in student reading performance accounted for by differences across teachers and schools. Variables at the student, teacher, and school levels were then entered step-by-step to develop the full model, resulting in three conditional models for each set of PVs (see Table 3). Both the estimated fixed and random effects were reported and interpreted. In addition, ICCs, the proportions of variances explained at each level (R^2 ; Hox et al., 2018; Snijders & Bosker, 2011), and standardized coefficients served as effect sizes (Ferron et al., 2008; Lora, 2018; Snijders & Bosker, 2011).

dataset were nested within the same schools, they were associated with more than one teacher, making nesting under one teacher not feasible. Thus, because teachers were nested within the same schools, we aggregated the teacher variables at the school level before conducting the analysis, leading to teacher and school variables being at the same level.

Results

Elastic net analysis and variable selection

Results of the Elastic Net analysis returned a list of variables ranked based on their relative statistical contribution to the variance of the outcome across levels for each PV. The results were not always consistent across PVs. For example, there were variables receiving high (top 20) rankings across all 10 sets of PVs (e.g., books at home); some variables, however, only made it into the rank on just one PV. Thus, as noted earlier, the final variable selection relied on a combination of criteria. These variables were identified by the Elastic Net analysis as statistically important in terms of their large coefficient magnitudes across PVs and/or theoretically supported. The selection process was accomplished by the panel of authors with corresponding expertise, by considering both the statistical and theoretical importance of the variables. For example, two of the student-level background variables—1) the number of books and 2) musical instruments at home—showed high rankings across nearly all the PVs. The grade repetition variable, however, was only flagged by the results of three PVs. After the examination of the variables' statistical and theoretical significance, the panel decided to include all three for further analysis. As illustrated by this example, the list of the variables was finalized after the agreement was achieved among the panelists. A total of 24, 22, and 22 variables at the student, teacher, and school levels were obtained (Appendix A1 - A3). We discuss these below.

Student-level variables

Aligned with previous studies, a number of background characteristics were strongly associated with reading achievement. These included: ISCED (educational) level, grade retention, books and musical instruments at home, and digital affordances (e.g., “Available for you to use at home: cell phone”). The most salient non-cognitive and metacognition variables included: assessing credibility, reading strategy, joy of reading, and academic aspiration. Peer support (e.g., “During the past 12 months, how often: Other students spread nasty rumors about me”) was identified as another potentially salient correlate of reading achievement. Finally, a number of variables capturing student perceptions of their teachers (e.g., “The teacher showed enjoyment in teaching”) emerged as potentially salient positive correlates of student reading achievement.

Teacher-level variables

A number of subjects in initial teacher preparation (e.g., religion/ethics, technology, social studies, arts) and the overlap between initial preparation and current teaching area (in such subjects as religion/ethics, science, technology, arts) emerged as potentially salient correlates at the teacher level. Furthermore, teacher employment, experience, strategies, beliefs about reading instruction, and reading habits were strongly associated with their students' reading performance. The reading-habit variable “I rarely or never read books,” for instance, was associated with lower levels of student

reading literacy. Lastly, job satisfaction (e.g., “The advantages of being a teacher clearly outweigh the disadvantages”) emerged as an important correlate.

School-level variables

First, school rural location and private ownership emerged as potentially the most important correlates at the school level. Student enrollment mattered in terms of admission recommendation by feeder schools, the number of girls, and the number of students with special needs. Three school resources and climate variables (e.g., truancy), one parental engagement variable (i.e., proportion of parents discussing their children’s progress with teachers), five school-use-of-assessment variables (e.g., assessment used to monitor yearly progress), and three extracurricular literacy variables (e.g., contributing to school newsletter or such out-of-school experiences as collaborating with a local newspaper) were selected. The next five selected variables focused on the school-level use of digital devices (e.g., adequate number of digital devices connected to the Internet) and career guidance.

Variable preparation for the MLM analysis

To prevent issues of model non-convergence, multicollinearity, and over-parameterization due to the large number of variables entered, conceptually similar variables were grouped and converted into composite variables (Appendix A1 – A3). The grouping process was grounded in literature and PISA recommendations for generating composite scale scores (OECD, 2019). When applicable, confirmatory factor analysis (CFA) was conducted for each scale as support of construct validation (i.e., evidence related to internal structure; AERA, APA, & NCME, 2014). The CFA model fit was deemed satisfactory in the presence of a non-significant χ^2 test ($p > .05$), the comparative fit index (CFI) and Tucker Lewis index (TLI) values over 0.90, and the root mean squared error of approximation (RMSEA) and standardized root mean square residual (SRMR) values below 0.08. The estimated factor scores were used as the composite variable.

The grouping process resulted in 17 grouped composite variables. Among these variables, 12 consisted of more than two items supporting CFA use, which was run using the R package **lavaan** (Rosseel et al., 2021). Results of the analyses supported acceptable model-data fit for these variables except for *perceptions of instructional support* (model did not converge), *teaching strategy* ($\chi^2 = 1129.94$, $df = 5$, $p < .001$, TLI = 0.60, CFI = 0.80, RMSEA = 0.28, SRMR = 0.10), and *use of digital devices* ($\chi^2 = 6.36$, $df = 2$, $p < .04$, TLI = 0.81, CFI = 0.94, RMSEA = 0.12, SRMR = 0.04). Details of the CFA results are included in the Appendix A5.

Following OECD (2019) recommendations, a principal component analysis (PCA) was applied if the CFA was not feasible (i.e., not identifiable CFA model due to only two indicators/items, poor CFA model fit) and the estimation of the first PC was used as the composite variable.³ The same procedure was applied by PISA to compute the

³ While CFA assumes common variance among the individual items that can be treated as factors/constructs, PCA analyzes the total variance and assumes that all the variance is attributed to components. That

ESCS index (OECD, 2019). In the current study, the PCA was conducted for the three variables with poor CFA model fit (i.e., *perceptions of instructional support*, *teaching strategy*, and *use of digital devices*) and the remaining five two-item variables. Results of the PCA revealed that the eigenvalues of the first component ranged from 1.05 (*peer interaction*) to 2.71 (*teaching strategy*; see Appendix A5) across the variables. Proportions of variance accounted for by the components ranged from 31.86% (*perceptions of instructional support*) to 78.65% (*school resources*).

The process resulted in a total number of 29 variables across levels, including 11, 8, and 11 variables at the student, teacher, and school levels, respectively. The variables consisted of original variables from PISA data (e.g., *ISCED level*), factor scores from CFA (e.g., *digital affordance*), and components from PCA (e.g., *effort*). Descriptive statistics of the variables are included in Table 4.

Multilevel modeling

The null model

The ICC estimated from the null model (Model 0) indicated that, for the entire student sample, there was 17.4% of the variance in student reading performance accounted for at the teacher and school levels. The ICCs for the ML and MN samples were 13.0% and 18.1%, respectively.

Multilevel models: entire sample

Table 5 presents the results of multilevel models for the entire student sample, including both standardized and unstandardized regression coefficients, and model fit indices. Specifically, Model 1 includes student-level variables only, Model 2 adds teacher-level variables, and Model 3 (the final model) includes all variables across levels. The likelihood ratio test (LRT) was used to compare the models. As indicated in the table, both the inclusion of student-level variables (Model 1 vs. Model 0; $F [11, 479] = 221.97, p < .001$) and school-level variables (Model 3 vs. Model 2; $F [10, 208,400] = 3.755, p < .001$) improved the model fit significantly. The teacher-level variables, however, did not show a significant contribution to the model, $F (7, 189,300) = 1.325, p = .233$.

Results of the final model served to interpret the relationship between selected variables and student reading performance on PISA for the entire student sample. R^2 statistics suggested that student-level variables accounted for 44% of the level 1 variance in the outcome while the teacher- and school-level variables accounted for 11% of the variance at level 2. All student-level variables, except for *words understood* ($b = 1.46, SE = 1.55, \beta = 0.01, p = .35$)⁴, showed a significant relationship with the outcome. Students' *grade repetition* showed an adverse effect on their reading per-

is, while CFA allows us to consider the factors/constructs underlying the data, the goal of PCA is to extract maximum variance from the data with components (Tabachnick & Fidell, 2013).

⁴ For ease of interpretation, we used b and β to represent unstandardized and standardized coefficients, respectively.

Table 4 Descriptive statistics of identified predictors

Variables	Entire Sample (<i>N</i> =4838)		Multilingual (<i>N</i> =736)		Monolingual (<i>N</i> =4054)	
	Mean	SD	Mean	SD	Mean	SD
Outcome						
Reading Achievement	500.15	108.45	480.63	105.71	504.66	108.32
Student-level predictors						
Effort [#]	0.00	1.01	-0.14	1.13	0.03	0.97
Words Understood (1 = Strongly disagree ~ 4 = Strongly agree)	1.87	0.76	1.89	0.75	1.86	0.76
ISCED Level (1 = Level 1 ~ 5 = Level 5)	2.92	0.28	2.91	0.29	2.92	0.28
Grade Repetition (0 = No, 1 = Yes)	0.10	0.30	0.13	0.34	0.10	0.29
Musical Instrument (1 = None ~ 4 = Three or more)	2.29	1.18	2.09	1.12	2.33	1.19
Books at Home (1 = 0–10 books ~ 6 = More than 500 books)	2.77	1.40	2.22	1.18	2.87	1.41
Digital Affordance [*]	0.00	0.85	0.01	0.85	-0.01	0.85
Metacognition [*]	0.00	0.89	-0.16	0.87	0.03	0.89
Self Efficacy [#]	-0.01	1.00	-0.11	1.03	0.01	0.99
Peer Interaction [#]	-0.01	1.00	-0.33	0.95	0.05	1.01
Perceptions of Instructional Support [#]	-0.01	1.00	-0.04	0.97	0.01	1.00
Teacher-level predictors						
Initial Training [*]	0.14	0.94	0.10	0.88	0.12	0.95
Training Relevance [*]	-0.10	0.67	-0.16	0.56	-0.08	0.68
Teaching Experience [*]	-0.05	1.45	-0.26	1.24	-0.05	1.33
Teacher Belief (1 = Strongly disagree ~ 4 = Strongly agree)	3.48	0.32	3.56	0.26	3.46	0.31
Teaching Strategy [#]	0.51	0.78	0.47	0.91	0.57	0.78
Reading Habit [#]	0.38	1.00	0.36	0.96	0.34	0.97
Job Satisfaction [*]	0.25	0.87	0.25	0.76	0.25	0.87
School-level predictors						
School Type (0 = Private, 1 = Public)	0.04	0.19	0.04	0.20	0.04	0.18
Enrollment [*]	-2.13	27.22	1.27	29.07	-1.93	26.29
School Resources [*]	-0.01	0.98	-0.19	0.98	0.05	1.00
Student Truancy (1 = Not at all ~ 4 = A lot)	2.78	0.83	2.66	0.80	2.79	0.85
Extracurricular Literacy [*]	-1.90	3.93	-2.51	3.22	-1.59	3.94
Discuss Progress with Parents (%)	50.47	24.52	52.32	24.06	52.10	25.98
Use of Assessment [*]	-1.47	3.52	-1.54	3.78	-1.57	3.62
Computer WWW Connection (Number of computers connected to WWW)	464.10	471.72	537.22	569.25	451.46	463.26
Use of Digital Devices [#]	-0.04	1.01	0.09	0.99	-0.07	1.04
Career Guidance Responsibility (0 = Not Checked, 1 = Checked)	0.03	0.16	0.03	0.16	0.03	0.16

Note. Construct/variables with * were latent factor scores obtained with confirmative factor analysis, while those with # were estimated scores of the first component obtained with the principal component analysis. Both the factor and component scores followed a standard normal distribution. Details of the questionnaire items consisting of the variables are included in Appendix Tables A1 – A3. The reading achievement was reported on a scale with a mean of 500 and a standard deviation of 100. ISCED=International Standard Classification of Education

Table 5 Model Parameter Estimates with the Entire Student Sample

Variable	Model 1			Model 2			Model 3		
	Est	SE	β	Est	SE	β	Est	SE	p
Fixed Parameters									
Intercept	501.06	3.91	0.00	501.19	3.89	0.01	501.41	3.58	<.001
Student-Level Predictors									
Effort#	12.19	1.22	0.11	12.20	1.22	0.11	12.20	1.22	<.001
Words Understood	1.45	1.55	0.01	1.45	1.55	0.01	1.46	1.55	.35
ISCED Level	33.64	5.43	0.09	33.63	5.43	0.09	33.64	5.43	<.001
Grade Repetition	-26.33	4.73	-0.07	-26.34	4.73	-0.07	-26.33	4.73	<.001
Musical Instrument	6.62	1.06	0.07	6.62	1.06	0.07	6.62	1.06	<.001
Books at Home	10.34	0.98	0.13	10.34	0.98	0.13	10.34	0.98	<.001
Digital Affordance*	26.23	1.48	0.21	26.23	1.48	0.21	26.23	1.48	<.001
Metacognition*	33.63	1.48	0.28	33.63	1.48	0.28	33.63	1.48	<.001
Self Efficacy#	15.57	1.24	0.14	15.57	1.24	0.14	15.57	1.24	<.001
Peer Interaction#	15.27	1.26	0.14	15.27	1.26	0.14	15.27	1.26	<.001
Perceptions of Instructional Support#	4.22	1.24	0.04	4.21	1.24	0.04	4.22	1.24	<.001
Teacher-Level Predictors									
Initial Training*	/	/	/	8.92	7.00	0.07	7.73	6.63	.24
Training Relevance*	/	/	/	-0.49	9.44	0.00	1.45	8.95	.87
Teaching Experience*	/	/	/	-4.00	2.90	-0.05	-3.25	2.71	.23
Teacher Belief	/	/	/	-17.51	12.35	-0.05	-25.40	12.00	.03
Teaching Strategy#	/	/	/	-4.84	5.37	-0.04	-1.28	5.14	.80
Reading Habit#	/	/	/	-2.37	4.18	-0.02	-0.95	3.95	.81
Job Satisfaction*	/	/	/	4.39	4.68	0.03	2.90	4.52	.52
School-Level Predictors									
School Type	/	/	/	/	/	/	-14.17	19.50	.47
(0= Private, 1= Public)									

Table 5 (continued)

Variable	Model 1 Student Predictors			Model 2 Student & Teacher Predictors			Model 3 Full Model		
	Est	SE	β	Est	SE	β	Est	SE	β
Enrollment*	/	/	/	/	/	/	-0.08	0.15	-0.02
School Resources*	/	/	/	/	/	/	-1.41	3.82	-0.01
Student Truancy	/	/	/	/	/	/	-24.64	4.67	-0.19
Extracurricular Literacy*	/	/	/	/	/	/	-1.23	1.00	-0.04
Discuss Progress with Parents (%)	/	/	/	/	/	/	0.11	0.15	0.03
Use of Assessment*	/	/	/	/	/	/	0.68	1.01	0.02
Computer WWW Connection	/	/	/	/	/	/	0.01	0.01	0.04
Use of Digital Devices*	/	/	/	/	/	/	-3.13	4.16	-0.03
Career Guidance Responsibility (0 = Not Checked, 1 = Checked)	/	/	/	/	/	/	4.11	24.06	0.01
Variance Components									
School (between)	2240.344			2213.838			1829.922		
Student (within)	5466.048			5465.598			5466.405		
Model Fit									
ICC	0.29			0.29			0.25		
Likelihood Ratio Test	$F(11, 479)=221.972, p<.001$			$F(7, 189300)=1.325, p=.233$			$F(10, 208400)=3.755, p<.001$		
L1 R Square	0.44			0.44			0.44		
L2 R Square	\			\			0.11		

Note. ICC = intraclass correlation Est = unstandardized coefficients, SE = standard errors, β = standardized coefficients.

formance ($b = -26.33$, $SE = 4.73$, $\beta = -0.07$, $p < .001$). All the other student-level variables revealed a positive relationship with the outcome, including *effort* ($b = 12.20$, $SE = 1.22$, $\beta = 0.11$, $p < .001$), *ISCED level* ($b = 33.64$, $SE = 5.43$, $\beta = 0.09$, $p < .001$), *musical instrument* ($b = 6.62$, $SE = 1.06$, $\beta = 0.07$, $p < .001$), *books at home* ($b = 10.34$, $SE = 0.98$, $\beta = 0.13$, $p < .001$), *digital affordance* ($b = 26.23$, $SE = 1.48$, $\beta = 0.21$, $p < .001$), *metacognition* ($b = 33.63$, $SE = 1.48$, $\beta = 0.28$, $p < .001$), *self-efficacy in reading* ($b = 15.57$, $SE = 1.24$, $\beta = 0.14$, $p < .001$), *peer interaction* ($b = 15.27$, $SE = 1.26$, $\beta = 0.14$, $p < .001$), and *perceptions of instructional support* ($b = 4.22$, $SE = 1.24$, $\beta = 0.04$, $p < .001$).

Relative to other predictors, there were only two variables at the teacher and school levels that showed a significant relationship with student reading performance. Specifically, *teacher belief* showed a significantly negative effect on the outcome, but only at a borderline level ($b = -25.40$, $SE = 12.00$, $\beta = -0.07$, $p = .03$). At the school level, the relationship between *student truancy* and the outcome was significantly negative ($b = -24.64$, $SE = 4.67$, $\beta = -0.19$, $p < .001$).

Multilevel models: multilingual vs. monolingual

Table 6 shows the results of the final model for the ML and MN student samples. Not surprisingly, the results of the MN sample are similar to those of the entire sample as the MN students account for the majority of the data (4054 out of the 4838 U.S. participants in PISA 2018). A comparison of the results for the two language group sub-samples, however, revealed different patterns. First, there are variables showing a significant relationship with the reading outcome for the MN, but not for the ML, students including *ISCED level* ($b = 38.53$, $SE = 6.14$, $\beta = 0.10$, $p < .001$), *perceptions of instructional support* ($b = 4.13$, $SE = 1.32$, $\beta = 0.04$, $p < .001$), and *teacher belief* ($b = -28.70$, $SE = 11.87$, $\beta = -0.08$, $p = .02$). None of these variables showed a significant effect on the reading performance of ML students.

Second, *extracurricular literacy* at the school level revealed a significantly positive relationship with the outcome only for ML students ($b = 3.24$, $SE = 1.56$, $\beta = 0.10$, $p = .04$). This variable was unique to the ML student sample. No significant association between *extracurricular literacy* and reading performance was detected neither for the ML nor for the entire student sample.

Third, examining the magnitudes of significant predictors showed that several associations between the predictors and the outcome were non-trivially stronger for ML students. For example, *grade repetition* had a substantially stronger negative impact on ML students reading achievement ($b = -42.54$ vs. -25.03 , $\beta = -0.14$ vs. -0.07) while higher levels of *self-efficacy* ($b = 20.63$ vs. 15.74 , $\beta = 0.20$ vs. 0.15) and stronger *peer interaction* ($b = 21.55$ vs. 15.26 , $\beta = 0.19$ vs. 0.14) tended to yield better performance for ML students in PISA reading. Overall, with the only exception of *digital affordance*, the magnitude of which was larger for MN students ($b = 26.10$ vs. 24.78 , $\beta = 0.20$ vs. 0.21), the associations between the statistically significant predictors and the reading outcome tended to be stronger for ML students.

Discussion

Reading literacy is foundational for full participation in contemporary society, including social, cultural, and political engagement (OECD, 2019). For instance, research links higher levels of reading literacy with reduced crime, better health, empowerment, and higher employability (Dewan, 2016; Morrisroe, 2014). PISA data represent a critical time period for fulfilling academic requirements for timely graduation and for setting foundations for future careers. This is particularly important for ML students, a group that is historically underrepresented in higher-paid jobs (Fraleigh-Lohrfink et al., 2013) and is still understudied, especially in higher grades (e.g., de Araujo et al., 2018; Hall et al., 2017). Thus, it is important to understand what factors impact reading literacy of ML students who navigate the schooling through an additional language while being held to the same accountability and academic standards as their MN peers. To address these gaps, the current study applied a three-step analytical strategy with support of previous empirical and theoretical research to identify the correlates of high school students' reading performance on the PISA 2018 assessment, with a particular emphasis on ML students.

From a methodological perspective, our study took advantage of a host of advanced statistical techniques to address our research questions. First, the Elastic Net approach, a popular machine learning variable selection technique, served to select the most salient factors associated with student reading (Research Question 1). The Elastic Net approach provides the advantages of efficient handling of a large number of variables (Chen et al., 2019; Wei et al., 2019), while working with large-scale data sets to identify variables with the highest contributions to student performance at all levels. Second, MLM served to explore and interpret the relationship between the selected factors and student reading performance at the student, teacher, and school levels (Research Question 2). In particular, the power of MLM analyses allowed us to evaluate—across nested levels of data—the relative contributions of the statistically significant variables. Finally, MLM analyses disaggregated by language status allowed us to test the generalizability of the obtained multilevel model for ML students, a student population at the heart of the present study (Research Question 3).

Expanding on this methodological discussion, we suggest that methods like Elastic Net serve a very important purpose in the field – that of theory generation and affirmation from an *atheoretical* perspective. We draw parallels to traditional approaches of grounded theory, wherein theories themselves are generated *from* the data. While grounded theory and large-scale assessments and surveys may at first glance seem to be a strange pairing, we echo Walsh et al., (2015) that grounded theory itself is not bound to any methodological approach, but rather a methodology that embraces any and all types of data. Glaser & Strauss (1967) make this point explicitly when arguing that grounded theory, at its core, is “derived from data and then illustrated by characteristic examples of data” (p. 5). If grounded theory is the “discovery of emerging patterns in data” (p. 593) as Glaser suggests (in Walsh et al., 2015), then applications of Elastic Net and other bottom-up approaches to data analysis may prove fruitful for theoretical validation and expansion as the field moves forward.

From a theoretical perspective, this study contributes to a limited body of work focusing on identifying correlates of high school students' reading achievement

Table 6 Full Model Parameter Estimates (Multilingual vs. Monolingual Students)

Variable	Multilingual Students (N=736)					Monolingual Students (N=4054)						
	Est	SE	β	t	df	p	Est	SE	β	t	df	p
Fixed Parameters												
Intercept	484.53	5.68	0.04	85.29	796	<.001	505.79	3.54	0.01	142.84	21658	<.001
<i>Student-Level Predictors</i>												
Effort	11.26	3.18	0.12	3.54	322	<.001	10.67	1.39	0.10	7.70	345	<.001
Words Understood	-2.40	4.48	-0.02	-0.54	541	.59	2.36	1.66	0.02	1.42	666	.16
ISCED Level	-8.70	13.97	-0.02	-0.62	505	.53	38.53	6.14	0.10	6.28	156	<.001
Grade Repetition	-42.54	12.05	-0.14	-3.53	440	<.001	-25.03	5.18	-0.07	-4.83	3227	<.001
Musical Instrument	8.76	3.37	0.09	2.60	280	<.001	6.42	1.15	0.07	5.57	1249	<.001
Books at Home	12.80	3.31	0.14	3.87	279	<.001	10.07	1.05	0.13	9.58	535	<.001
Digital Affordance	24.78	4.08	0.20	6.08	1318	<.001	26.10	1.65	0.21	15.85	390	<.001
Metacognition	33.96	4.01	0.28	8.47	1927	<.001	33.48	1.58	0.28	21.24	522	<.001
Self Efficacy	20.63	3.28	0.20	6.28	2975	<.001	15.74	1.34	0.15	11.73	1055	<.001
Peer Interaction#	21.55	3.80	0.19	5.67	1208	<.001	15.26	1.34	0.14	11.39	299	<.001
Perceptions of Instructional Support	5.11	3.62	0.05	1.41	315	.16	4.13	1.32	0.04	3.13	319	<.001
<i>Teacher-Level Predictors</i>												
Initial Training	-1.68	10.38	-0.01	-0.16	4407	.87	7.58	6.64	0.06	1.14	56130	.25
Training Relevance	6.94	14.91	0.04	0.47	2810	.64	5.61	9.06	0.04	0.62	37715	.54
Teaching Experience	-6.63	4.37	-0.08	-1.52	856	.13	-3.03	2.78	-0.04	-1.09	23500	.28
Teacher Belief	1.32	20.17	0.00	0.07	5016	.95	-28.70	11.87	-0.08	-2.42	28257	.02
Teaching Strategy	8.71	7.29	0.08	1.19	2429	.23	-3.13	5.42	-0.02	-0.58	414915	.56
Reading Habit	-9.89	6.72	-0.09	-1.47	6714	.14	-1.87	4.01	-0.02	-0.47	356220	.64
Job Satisfaction	-6.24	7.71	-0.05	-0.81	2423	.42	2.42	4.43	0.02	0.55	17045	.58
<i>School-Level Predictors</i>												
School Type (0=Private, 1=Public)	-25.22	28.79	-0.05	-0.88	8302	.38	-12.59	19.37	-0.02	-0.65	5031	.52
Enrollment	-0.03	0.23	-0.01	-0.14	1226	.89	0.02	0.15	0.01	0.15	11980	.88

Table 6 (continued)

Variable	Multilingual Students (N=736)						Monolingual Students (N=4054)					
	Est	SE	β	<i>t</i>	<i>df</i>	<i>p</i>	Est	SE	β	<i>t</i>	<i>df</i>	<i>p</i>
School Resources	1.99	6.07	0.02	0.33	14700	.74	-4.97	4.05	-0.05	-1.23	77545	.22
Student Truancy	-25.28	8.07	-0.17	-3.13	717	<.001	-23.08	4.46	-0.20	-5.18	15874	<.001
Extracurricular Literacy	3.24	1.56	0.10	2.08	8222	.04	-1.03	0.95	-0.04	-1.09	24585	.28
Discuss Progress with Parents (%)	0.20	0.25	0.05	0.78	3137	.44	0.02	0.15	0.00	0.13	35996	.89
Use of Assessment	1.13	1.61	0.04	0.70	1727	.48	-0.03	1.04	0.00	-0.03	19270	.97
Computer WWW Connection	0.00	0.01	0.01	0.20	186746	.84	0.01	0.01	0.03	0.93	11608	.35
Use of Digital Devices	0.75	6.36	0.01	0.12	3579	.91	-0.91	4.06	-0.01	-0.22	11462	.82
Career Guidance Responsibility (0 = Not Checked, 1 = Checked)	10.72	29.22	0.02	0.37	2240	.71	36.35	23.17	0.06	1.57	8244	.12
Variance Components												
School (between)	1963.9						1727.0					
Student (within)	5532.3						5356.6					
Model Fit												
ICC	0.26						0.24					
Likelihood Ratio Test	<i>F</i> (10, 761)=1.851, <i>p</i> =.049						<i>F</i> (10, 193481)=4.85, <i>p</i> <.001					
L1 R Square	0.44						0.44					
L2 R Square	\						0.19					

Note. ICC = intraclass correlation, Est = unstandardized coefficients, SE = standard errors, β = standardized coefficients.

across multiple nested systems. Consistent with Bronfenbrenner's (1977) ecological framework, the results identified a number of variables at each level impacting students' reading achievement. This finding is also consistent with the well-recognized and commonly implemented approach in educational research examining the impacts of multiple levels within school systems. In turn, using machine learning approaches and testing model generalizability for ML vs. MN student populations, this study also contributes to generating insights that may lead to inductive theory building (Choudhury et al., 2018). Future studies will need to address the limitations of PISA data to capture the time and interaction components of Bronfenbrenner's (1977) model.

In particular, the results of the Elastic Net analysis, supported by the literature, identified a total of 68 original variables in the PISA data as the most salient correlates of reading performance across levels (24 at the student and 22 at each the teacher and the school levels). After grouping and constructing composite variables, their relative associations with reading were explored using MLM. Results for the combined sample indicated that most of the student-level correlates were significant, including both negative and positive correlates. The former included *grade repetition*; the latter included *effort*, *ISCED level*, *musical instrument*, *books at home*, *digital affordance*, *metacognition*, *self-efficacy in reading*, *peer interaction*, and *perceptions of instructional support*.

At the teacher and school levels, there were only two significant correlates. Consistent with both theory (Maslow, 1943) and previous literature (Astor et al., 2009), the one at the school level, *student truancy*, was a negative correlate of student achievement. Surprisingly, *teacher belief*, a teacher-level correlate, was also negative for the combined and MN samples, perhaps due to reflecting teachers' aspirations rather than their actual training. This, in combination with current curriculum for adolescents often not including direct reading instruction (OECD, 2019) and the fact that only about 30% of teachers in the PISA sample were reading or language-specific teachers, who, by definition, are trained to teach reading comprehension, may explain this unexpected finding. More research is needed to explore the impacts of content-area teachers' actual training to teach reading comprehension, especially for ML students in high school where the language demands of schooling are typically higher. Finally, as we discuss next, the examination of the model generalizability revealed additional differences in model performances for ML students.

From a practical perspective, some background variables at the student level identified as salient and statistically significant correlates of high school readers' performance are outside of direct impacts of educational systems (e.g., books and musical instruments at home), unless through parent education, outreach, and financial support programs. That said, a host of student-level variables related to reading performance that remained significant when considering the combined influences at all levels are manipulable by instruction and/or larger educational systems (Table 4). Most pertinent to ML students are *grade retention*, *peer interaction*, *perceptions of instructional support*, and *reading self-efficacy*. In particular, unlike for the combined or the MN samples, *grade retention* was the strongest and negative correlate of high school ML student reading performance. This is of concern as research suggests that disparities associated with school punishment practices and academic progress for ML students persist, especially in urban schools (Peguero et al., 2021). Substantively

larger associations with *peer interaction* for ML as compared to MN students – captured by two items in our study, namely, language usually spoken with best friend and frequency of bullying (“nasty rumors”) behaviors – call for more institutional attention focused on properly fostering this social support factor, especially for ML students. In particular, schools need to strive for greater diversity in their enrollment as languages usually spoken with best friends was one of the indicators of positive peer relationships and there is a substantial body of research supporting the positive role of native language support in academic development, including developing supportive networks and resiliency to adverse factors (e.g., Stevenson et al., 2019). Having peers sharing the same language in the school may also minimize bullying behaviors, provided that schools implement anti-bullying initiatives at the individual, peer-group, and school-wide levels (Carney et al., 2018; Haeseler, 2010; Salmivalli, 2010). Substantively larger associations with *reading self-efficacy*, however, do not necessarily suggest that an instructional focus on this variable is needed given that our findings regarding self-efficacy impacts may be an artifact of the cross-sectional nature of the data and some recent findings suggesting that reading achievement influences reading self-efficacy over time, but the opposite influence may be relatively limited (e.g., Burns et al., 2020; Sewasew & Koester, 2019).

A unique lack of a statistically significant association between ML student achievement and *perceptions of instructional support*, operationalized in our study as student perceptions of teacher instructional support (e.g., amount of assigned readings, teacher summarizing previous lessons or inviting students’ sharing favorite passages) is surprising. This lack of significant association may be attributed to teachers generally being (under)prepared to meet the needs of their diverse learners (Butcher, 2020; Okhremtchouk & Sellu, 2019; Samson & Collins, 2012), thus likely providing support not targeted to meet the learning needs of ML learners. For example, inviting publicly sharing a passage without rehearsal (e.g., turn and talk, think-pair-share), may increase levels of anxiety, especially if facing unsupportive peers, and undermine active participation of ML with still-developing language skills. Somewhat relatedly, *extracurricular literacy* (additional lectures/seminars, contributing to school newsletter, collaborating with a local newspaper) was a unique contributor to ML student reading comprehension, plausibly reflecting ML students’ need for more meaningful, real-life applications of English literacy skills and additional exposure to academic English outside of school.

Study limitations and future directions

Every study has limitations, and this one is no different. First, the amount of unexplained variance in reading achievement suggests that variables other than those included in the present study may need to be included in future investigations. One such possible variable is a more nuanced conceptualization of the language proficiency for ML learners, as the needs of English language beginners (the most vulnerable ML student population in upper grades) may differ from students who are early intermediate, intermediate, or early advanced speakers and readers of English. It may be beneficial to collect such information for future PISA assessment administrations, especially for such subjects as literacy. Further, although ML status—also

often referred to in the literature as limited English proficiency, English learner, or language minority status—is often associated with lower SES, cultural/ethnic diversity, and overrepresentation in students with disability status (e.g., Becker & Deris 2019; Lobstein et al., 2021), potentially leading to collinearity issues, it may be helpful to separately examine the contributions of these demographic variables to reading achievement of MLs, both at the student and classroom or school levels. Moderator analysis will likely also provide a fruitful path for future research especially to help explain the remaining variance in reading test scores. Economic indicators, such as ESCS, PISA's indicator of economic, social, and cultural status, have been used in prior research as within-level and cross-level moderators and should be explored in the context of ML student research as well in the future.

The second limitation lies in the large number of variables put forward by the Elastic Net approach and the potential concerns of multicollinearity, model non-convergence, and over-parameterization in statistical modeling. An advantage of Elastic Net (and other approaches with similar purposes such as *Lasso*, *Ridge*, and *Tree-based methods*) is to assist in the selection of variables with high statistical contributions to the outcome in the presence of a large p (number of variables) and a small n (sample size). It is possible that the final selected list is still too large and includes similar variables, especially in the context of large-scale assessments and surveys where plausible values are used. To address this issue, two options can be used. One is to combine the variables into composites or indices, while the other is to exclude variables that have high correlations with other variables (e.g., OECD, 2008). In our study, we followed the first option by applying CFA and PCA techniques. While most of the grouped variables were supported either as a factor with a satisfactory CFA model data fit or a component that accounted for a large proportion of variance in PCA, we noticed specific variables that need further investigation. For instance, the *perceptions of instructional support* component (discussed earlier) only accounted for about 32% of the variance.

Third, concerns may lie in the sample size differences across the compared student subsamples (4,054 MNs from 163 schools vs. 736 MLs from 126 schools). While the sample sizes at both levels are acceptable for the multilevel modeling analysis (Hox et al., 2018; Snijders & Bosker, 2011), issues may arise when the coefficients and effects across the two student samples are compared to each other. For instance, there were three variables (i.e., *ISCED level*, *perceptions of instructional support*, and *teacher belief*) that showed significant effects on reading literacy for the MN students only. This difference could potentially be attributed to the smaller sample size of MLs. Therefore, as recommended in the literature, we examined the effect sizes in addition to the significance levels of the coefficients. For instance, the variable *perceptions of instructional support* was a significant predictor of reading performance for MNs ($b=4.13$, $SE=1.32$, $\beta=0.04$, $p<.001$), but not for MLs ($b=5.11$, $SE=3.62$, $\beta=0.05$, $p=.16$). However, the very similar size in corresponding standardized coefficients, $\beta=0.04$ and $\beta=0.05$, respectively, suggest that the *perceptions of instructional support* variable may also have a statistically significant effect on MLs' reading literacy, if the ML sample were larger.

To further investigate this issue and inform future studies, we conducted a follow-up analysis using a random sample of MN students with the same number of schools

and students as that of the ML sample. First, we drew a random sample of 126 schools from the MN sample (the same number of schools in the ML sample). Then, we drew a random sample of 736 MN (matching the ML size). The same multilevel modeling analysis was then conducted using the reduced-size sample of MN students. Results of the reduced subsample analysis supported our full sample analysis conclusions, on all but one of the variables (see Appendix Table A6). The *teacher belief* variable became not significant, $p = .32$, in the reduced sample (vs. $p = .02$ in the full sample) and the magnitude of the effect size also slightly decreased from $\beta = 0.08$ in the full to 0.06 in the reduced sample.

Finally, due to the cross-sectional nature of PISA data, we were not able to explore the impacts of variables over time, which is important from both theoretical (e.g., single episodes vs. systematic reoccurrence; Bronfenbrenner & Morris 1998) and practical (e.g., informing intervention duration) perspectives.

Conclusion

The results of our study echo previous literature in many respects. The findings also identify a host of correlates at each level not investigated in the previous research and highlight their relative, statistically significant associations with reading. The multistep approach used in this study both identifying the most strongly associated correlates of reading and then examining their relative associations may shed light on the existing literature from a new angle and thus contribute to refining existing theories. The use of machine learning, in particular, which is a largely data-driven approach with few a priori assumptions, offers a practical application of a methodological innovation—still novel in many areas of educational research—different from many studies in the field that focus on hypothesis testing. As such, the study, as pointed out by one of the anonymous reviewers, can serve as a way to validate—and expand upon—existing findings from theory-based studies.

From a practical perspective, the results suggest that ML students would especially benefit from increased learning opportunities for extracurricular literacy activities. From a policy perspective, the results suggest that all students, but especially ML students, would benefit from schools avoiding grade retention policies and focusing on promoting a school climate supporting positive peer relationships and tailored teacher practices, which should also minimize truancy. From a future research perspective, it is paramount to focus more attention on high school ML students, a burgeoning and often at-risk student population in our schools. Research is needed that considers the impacts of multiple, nested systems on improving reading literacy for ML students, especially considering the high cost of interventions and their differential impacts on ML versus MN students.

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